

Mock Exam

Answering Sheet

First Name, Surname: _____

Study Program: _____

Signature: _____

Student-ID: 0 _____

0	☒	☐	☐	☐	☐	☐	☐	☐
1	☐	☐	☐	☐	☐	☐	☐	☐
2	☐	☐	☐	☐	☐	☐	☐	☐
3	☐	☐	☐	☐	☐	☐	☐	☐
4	☐	☐	☐	☐	☐	☐	☐	☐
5	☐	☐	☐	☐	☐	☐	☐	☐
6	☐	☐	☐	☐	☐	☐	☐	☐
7	☐	☐	☐	☐	☐	☐	☐	☐
8	☐	☐	☐	☐	☐	☐	☐	☐
9	☐	☐	☐	☐	☐	☐	☐	☐

Example on how to fill out:

Picking Option B: ☐ A ☒ B ☐ C ☐ D

Changing Answer to Option D: ☐ A ☒ B ☐ C ☒ D

Q	Section 1.1
1	☐ A ☐ B ☐ C ☐ D
2	☐ A ☐ B ☐ C ☐ D
Q	Section 1.2
3	☐ A ☐ B ☐ C ☐ D
4	☐ A ☐ B ☐ C ☐ D
Q	Section 1.3
5	☐ A ☐ B ☐ C ☐ D
Q	Section 1.4
6	☐ A ☐ B ☐ C ☐ D
7	☐ A ☐ B ☐ C ☐ D
Q	Section 1.5
8	☐ A ☐ B ☐ C ☐ D

9	☐ A ☐ B ☐ C ☐ D
Q	Section 2.1
10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D
12	☐ A ☐ B ☐ C ☐ D
Q	Section 2.2
13	☐ A ☐ B ☐ C ☐ D
14	☐ A ☐ B ☐ C ☐ D
Q	Section 2.3
15	☐ A ☐ B ☐ C ☐ D
Q	Section 2.4
16	☐ A ☐ B ☐ C ☐ D
17	☐ A ☐ B ☐ C ☐ D

18	☐ A ☐ B ☐ C ☐ D
Q	Section 2.5
19	☐ A ☐ B ☐ C ☐ D
20	☐ A ☐ B ☐ C ☐ D
Q	Section 2.6
21	☐ A ☐ B ☐ C ☐ D

Instructions:

- Total processing time: 120 minutes (no extra time)
- Total points: 120 (it should take about one minute to complete one point)
- **Only answers given on the answering sheet will be considered.**
- For each question, only one answer will be correct. Mark the correct answer on the answering sheet with a dark pen.
- **No** negative points will be assigned.
- Do not unstaple the question sheets or detach pages from this exam. Otherwise, your exam will not be evaluated.
- Please do not round intermediate results.
- Throughout the exam, commas are used to separate thousands and dots are used to separate decimals, i.e. 10,000 EUR means ten thousand euros and 10.40 EUR means ten euros and forty cents.
- Please verify the completeness of your exam.
- If you need extra space, use the backsides of each page.
- Only the following resources are allowed to be used:
 - Dark pen,
 - Non-electronic dictionary,
 - Non-graphical and non-programmable calculator.

Good luck!

Section 1: Lecture

Section 1.1: Financial Analysis (4 points)

Question 1

Evaluate the following statements.

- (1) Financial statements are e.g. analyst-based reports including future performance information.
- (2) Financial statements are prepared in accordance with generally accepted accounting principles or reporting standards to ensure consistency and comparability.
- (3) Financial statements are relatively unreliable sources of information for decision-making, financial analysis, and regulatory compliance.

- A) (1) is correct. (2) and (3) are false.
- B) (2) is correct. (1) and (3) are false.
- C) (1) and (2) are correct. (3) is false.
- D) (2) and (3) are correct. (1) is false.

(2 points)

Question 2

The following information is given for the Fintech AI company:

Item	Value
Book Value of Equity	550 million EUR
Net Profit Margin	12%
Interest Expense	150 million EUR
Sales	900 million EUR

Based on the information above, what are the Fintech AI's Net Income and Interest Coverage Ratio (EBIT / Interest Expense)?

- A) Net Income = 108 million EUR; Interest Coverage = 1.72
- B) Net Income = 108 million EUR; Interest Coverage = 2.0
- C) Net Income = 120 million EUR; Interest Coverage = 1.8
- D) Net Income = 135 million EUR; Interest Coverage = 2.1

(2 points)

Section 1.2: Investment Analysis (5 points)

Question 3

Consider the following mutually exclusive projects.

Project	Period 0	Period 1	Period 2	∞
A	-75,000 €	38,000 €	40,000 €	
B	-90,000 €	55,000 €	45,000 €	
C	-100,000 €	8,500 €	8,500 €	8,500 €

Which of the following statements is **false**?

- A) Applying the NPV rule, you would never invest in Project A if the cost of capital is 4%.
- B) Applying the IRR rule, you would always turn down Project A if the cost of capital is at 2%.
- C) Applying the NPV rule, Project C is more attractive than Project B at a cost of capital of 6%.
- D) Applying the IRR rule, you would take Project C if the cost of capital is at 7%.

(3 points)

Question 4

Which of the following statements is **false**?

- A) The IRR can be used to check the NPV for any estimation error in the cost of capital.
- B) For mutually exclusive projects, you would always select the project with the highest IRR.
- C) The IRR cannot be used to compare projects of different scales.
- D) None of the above statements is false.

(2 points)

Section 1.3: Capital Budgeting (3 points)

Question 5

A company is evaluating a 3-year investment project. The project requires an initial investment of €150,000 in equipment (to be depreciated straight-line to zero over 3 years) and will generate additional revenues of €100,000 and cash operating costs of €40,000 per year. The tax rate is 30%, and there is no salvage value at the end of the project. What is the annual free cash flow (FCF) from this project?

- A) 42,000 EUR
- B) 50,000 EUR
- C) 57,000 EUR
- D) 60,000 EUR

(3 points)

Section 1.4: Cost of Capital Estimation (5 points)

Question 6

In mid-2012, Cisco Systems had a market capitalization of 98 million EUR. It had A-rated debt of 40 million EUR as well as cash and short-term investments of 10 million EUR, and its estimated equity beta at the time was 1.16. What is Cisco's debt beta if we assume unlevered (i.e., asset) beta of 1.0.

- A) 0.48
- B) 0.83
- C) 0.36
- D) 0.17

(3 points)

Question 7

You want to assess the beta, i.e. the sensitivity to the market portfolio, of a company that is not publicly traded. To do so, which is the correct order of the following steps:

- (1) Lever the beta for the company's financial risk.
- (2) Unlever the comparable company's beta to estimate the asset beta.
- (3) Select a comparable company.
- (4) Estimate the beta for the comparable company.

- A) (1) → (2) → (4) → (3)
- B) (3) → (4) → (2) → (1)
- C) (3) → (4) → (1) → (2)
- D) (3) → (1) → (2) → (4)

(2 points)

Section 1.5: Capital Structure (5 points)

Question 8

Modigliani and Miller propose the following equation (MM proposition II):

$$r_e = r_u + \frac{D}{E} \times (r_u - r_D)$$

Which relation does this equation describe?

- A) If a firm is levered, all of the free cash flows generated by its assets are paid out to its equity holders.
- B) The cost of capital of unlevered equity is equal to the cost of capital of levered equity plus a premium that is proportional to the market value of the debt-to-equity ratio.
- C) The market value, risk, and cost of capital for the firm's assets and its debt coincide.
- D) None of the above options describes the equation.

(2 points)

Question 9

NextGen Corp. has an equity beta of 1.50 and a debt-to-equity ratio of 0.40. Estimate its asset beta assuming the debt beta is zero. Then, suppose NextGen increases its leverage so that the new debt-to-equity ratio becomes 0.60. Assuming the debt beta remains zero and there are no taxes, what would be the new equity beta after the increase in leverage?

- A) Asset beta = 1.07; New equity beta = 1.54
- B) Asset beta = 1.25; New equity beta = 1.54
- C) Asset beta = 1.07; New equity beta = 1.71
- D) Asset beta = 1.25; New equity beta = 1.71

(3 points)

Section 2: Exercise

Section 2.1: Interest (7 points)

Question 10

The key interest rates in Euro-Zone can be subdivided into:

- A) Deposit facility, main refinancing rate, and Libor.
- B) Deposit facility, main refinancing rate, and marginal lending rate.
- C) Euribor, main refinancing rate, and marginal lending rate.
- D) Borrowing rate, interest on credit, and marginal lending rate.

(2 points)

Question 11

At an annual nominal interest rate of 3% with compounding interest, how many years will it take for an investment to double?

- A) 21.34 years
- B) 23.45 years
- C) 25.41 years
- D) 27.82 years

(2 points)

Question 12

An investor invests 1,000 EUR today. Her investment goal is to increase her initial investment by 2,000 EUR. With continuous compounding interest rates, the investor reaches her goal 30 years earlier compared to using simple interest calculation (without compounding). Which is the combination of N (with compounding) and interest rate that can achieve this?

- A) 37.8 years and 2.95%
- B) 40.1 years and 2.13%
- C) 36.6 years and 3.00%
- D) 39.2 years and 2.73%

(3 points)

Section 2.2: Annuities (6 points)

Question 13

A student regularly deposits 200 EUR at the beginning of each month into a savings account that earns monthly compounding interest at a nominal annual interest rate of 5.2%. What will be the account balance after 6 years??

- A) 19,428 EUR
- B) 18,889 EUR
- C) 16,930 EUR
- D) 15,960 EUR

(3 points)

Question 14

A student wants to regularly pay at the beginning of each month 100 EUR into a savings account for the next 8 years to be able to deposit their household after the studies. Interest rate is calculated on a monthly compounding basis with a nominal annual interest rate of 4.5%. What is the present value of this savings goal?

- A) 8,080 EUR
- B) 9,090 EUR
- C) 10,000 EUR
- D) 10,960 EUR

(3 points)

Section 2.3: Redemption (3 points)

Question 15

At the beginning of the second year, a student receives a loan of €120,000 to finance their studies. The degree program lasts 3 years. The student does not plan to repay any principal during the study period but considers paying interest during the studies in order to reduce the total repayment time after graduation. After graduation, the loan is repaid through annual annuity payments of €12,000, starting one year after graduation. The annual interest rate is 5.5%, and it is assumed to remain constant over the entire repayment period. How many years earlier can the student repay the loan if they choose to pay interest during the study period compared to deferring all payments until after graduation?

- A) 2.78 years
- B) 3.25 years
- C) 4.52 years
- D) 5.78 years

(3 points)

Section 2.4: Bonds (6 points)

Question 16

Which event does **not** have an immediate effect on the market prices of bonds?

- A) A new credit risk assessment by the market.
- B) A coupon is paid.
- C) The issuer declares bankruptcy.
- D) A change in the curvature of the yield curve.

(2 points)

Question 17

An investor purchases a long-term zerobond and holds it until maturity. Which of the following risks does the investor avoid?

- A) Default risk
- B) Reinvestment risk
- C) Interest rate risk
- D) All of the above

(2 points)

Question 18

A coupon bond with a Macaulay Duration of 4.79 years is trading at 95.46 EUR. Estimate the new bond price if the market interest rate decreases from 6% to 3%.

- A) 108.78 EUR
- B) 107.99 EUR
- C) 108.40 EUR
- D) 108.22 EUR

(2 points)

Section 2.5: Stocks (5 points)

Question 19

Virgo Airlines will pay a 4 EUR dividend next year on its common stock, which is currently selling at 100 EUR per share. What is the market's required return on this investment if the dividend is expected to grow at 5% forever?

- A) 4%
- B) 5%
- C) 7%
- D) 9%

(3 points)

Question 20

Which of the following statements is correct without limitations?

- A) Common stockholders have a senior claim on a company's assets and are paid before bondholders and preferred shareholders in the event of liquidation.
- B) A stock's market price always equals its intrinsic value, as calculated by a discounted cash flow model.
- C) If a company consistently pays high dividends, it always means the stock is a good investment.
- D) The dividend discount model assumes a constant growth rate in dividends and is best suited for valuing mature, stable companies.

(2 points)

Section 2.6: Options (2 points)

Question 21

An investor sells a European call option for 2.50 EUR. The current stock price is 41 EUR, and the strike price of the option is 50 EUR. Which of the following is correct without limitations?

- A) The option buyer generates a profit if the stock price is below 41 EUR upon expiration.
- B) The option buyer generates a profit if the stock price is between 41 EUR and 50 EUR upon expiration.
- C) The option seller generates a profit if the stock price is below 52.50 EUR upon expiration.
- D) The option seller generates a profit if the stock price is above 52.50 EUR upon expiration.

(2 points)

End of the exam!